■# PAPER<i>199: F</i>UBii Taxonomy Part 2 — Cosmological and Dark Sector Buoyancy Forces

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__Session:__ 50 — grokshare7514fe.txt Full Audit

■# PAPER*199: F*UBii Taxonomy Part 2 — Cosmological and Dark Sector Buoyancy Forces

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```
F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa =
5.0\times10^{-4}\,\text{day}^{-1},\; [SSq] = 0.57
```

```
\rho_\Lambda^\text{UQFF} = \rho_\Lambda^\text{obs}\cdot\Bigl(1 +
\kappa^2\cdot[SSq]^2\Bigr) = \rho_\Lambda^\text{obs}\times1.0000000812
```

<!-- ? = 5.0e-4 day?[1], [SSq] = 0.57, ß_i = 6.1e-1 -->

Abstract

This paper catalogs the second major group of F*UBii variants from the BB*C*Equations*04Sept2025.pdf (177-page equation catalogue): cosmological and dark sector buoyancy forces. Covered are anyon systems, dark energy, inflation, gravitational waves, Loop Quantum Cosmology (LQC) bounce and perturbation spectra, Bekenstein-Hawking entropy, Hawking evaporation lifetime, reheating, Big Bang Nucleosynthesis, reionization, baryon-photon ratio, convective turnover time, CMB angular power spectrum, recombination, NFW dark matter profiles, SIDM core formation, void density evolution, and peculiar velocity. Each F*UBii,X form uses the universal F*rel/E*LEP scaling with a system-specific physical expression F*X.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?[1], [SSq] = 0.57) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. Dark Sector and Quantum Gravity Variants

1.1 Dark Energy Buoyancy

```
F_UBii,DE = -F_rel × (?_DE·c² / E_LEP) × Q_wave(z) × (8pG?_tot/3) × (1+w(z))
?_DE(a) = ?_DE,0·exp(3?1^a (1+w(a'))/a' da')
w(a) = w0 + w_a(1-a) (Chevallier-Polarski-Linder parametrization)
Source: BB_C_Equations item 721
```

1.2 Inflation Buoyancy

```
F_UBii,inf = F_rel × (V(?) / E_LEP) × Q_wave × 3H² × e^N/(1+e)
V(?) = inflaton potential
N = number of e-folds
e = slow-roll parameter: e = (??/H·M_pl)²/2
Source: BB_C_Equations item 724
```

1.3 Gravitational Wave Energy Density Buoyancy

```
F_UBii,GW = -F_rel × (?_GW / E_LEP) × Q_wave × (32pG?²/c²) × e^{-t/t}
?_GW = gravitational wave energy density
? = time derivative of strain
Source: BB_C_Equations item 727
```

1.4 Anyon System Buoyancy

```
F_UBii,anyons = -F_rel × (E_anyons / E_LEP) × Q_wave × g(r,t) × exp(-d²_c/(2s²))
E_anyons = anyon system energy (2D topological)
g(r,t) = UQFF gravitational field at position
Gaussian factor from density fluctuation s
Source: BB_C_Equations item 710
```

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2. Loop Quantum Cosmology (LQC) Variants

2.1 LQC Perturbation Spectrum Buoyancy

```
F_UBii,lqcp = -F_rel × (P(k) ? k^{n_s-1}·(1+k/k_*)^{-a} / E_LEP) × Q_wave
k_* = bounce scale (LQC pre-bounce phase)
a = UV suppression exponent
Modifies primordial power spectrum at Planck-scale modes
Source: BB_C_Equations item 1431, 1254
```

2.2 LQC Effective Friedmann Buoyancy

```
F_UBii,lqcf = F_rel × (H² = 8pG?/3·(1-?/?_crit) / E_LEP) × Q_wave
?_crit = 0.41·?_Planck ˜ 10?³ g/cm³
Bounce condition: H=0 at ?=?_crit (avoids singularity)
Source: BB_C_Equations item 1252, 1604
```

2.3 LQC Bounce Timescale Buoyancy

```
F_UBii,bounc = F_rel × (t_b ˜ v(3/(8pG?_crit)) ˜ t_Pl ˜ 10?4³ s / E_LEP)
× Q_wave × [H˜0 at bounce, duration ˜ 1/?_bounce]
Source: BB_C_Equations item 1257, post-1608
```

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3. Black Hole Thermodynamics Variants

3.1 Bekenstein-Hawking Entropy Buoyancy

```
F_UBii,bhent = F_rel × (S = k_B·c³·A/(4Gh) = 4p·k_B·GM²/(hc) / E_LEP) × Q_wave
A = 4pr_s² (Schwarzschild horizon area)
r_s = 2GM/c²
Holographic: S ? A/l_Pl² (area law)
Source: BB_C_Equations item 1092, 1248
```

3.2 Evaporation Lifetime Buoyancy

```
F_UBii,evapl = -F_rel × (t_evap = 5120pG²M³/(hc4) / E_LEP)
× Q_wave × [P = sAT4 ˜ hc²/M²]
Power dM/dt = -P/c² (mass loss rate)
Source: BB_C_Equations item 1601, 1250
```

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4. Cosmological Nucleosynthesis and Reionization Variants

4.1 Big Bang Nucleosynthesis Deuterium Bottleneck

```
F_UBii,deb = -F_rel × (t_D = v(3/(32pG?_rad)) ˜ 180 s (T~0.1 MeV) / E_LEP) × Q_wave
?_rad = p²kT4/(30h³c5)·g_* (radiation density)
g_* = effective degrees of freedom (~10 at D formation)
Weak freeze-out at T~1 MeV; photodissociation until T<0.1 MeV
Source: BB_C_Equations item 1809, 1536
```

4.2 Baryon-to-Photon Ratio Buoyancy

```
F_UBii,eta = F_rel × (? = n_b/n_? = 6×10?¹° / E_LEP) × Q_wave × [Freeze-out: R...]
n_? = 410 cm?³ (CMB photon density today)
Fit from D, ³He, 4He, 7Li abundances
Source: BB_C_Equations item 1701, 1534
```

4.3 Reionization Front Buoyancy

```
F_UBii,reion = F_rel × (d?_e/dt = ?_?·e_esc·f_* - a_B·n²_e·C / E_LEP) × Q_wave
x_e = ionized fraction, e_esc ˜ 0.1–0.3 (escape fraction)
C = clumping factor
Source: BB_C_Equations item 1684
```

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5. CMB and Recombination Variants

5.1 CMB Angular Power Spectrum Buoyancy

```
F_UBii,cmb = F_rel × (C_l = 2/p · ?k²dk·P(k)·|?_l^T(k)|² / E_LEP) × Q_wave
P(k) ? k^{n_s-4} (primordial power, n_s ˜ 0.965)
Transfer ?_l^T: Sachs-Wolfe (large scales) + acoustic (small scales)
Source: BB_C_Equations item 1310, 1080
```

5.2 Recombination Optical Depth Buoyancy

```
F_UBii,recomb = -F_rel × (t(z) = ?_z^8 n_e(z')·s_T·c·(dt/dz')dz' / E_LEP) × Q_wave
z_re ˜ 7.7 (reionization redshift, Planck 2018)
s_T = Thomson cross-section (6.65×10?²? m²)
Source: BB_C_Equations item 1313
```

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6. Dark Matter Variants

6.1 NFW Density Profile Buoyancy

```
F_UBii,nfw = -F_rel × (?(r) = ?_s/((r/r_s)·(1+r/r_s)²) / E_LEP) × Q_wave
?_s = characteristic density, r_s = scale radius
Universal CDM halo form
Source: BB_C_Equations item 1326
```

6.2 NFW Rotation Curve Buoyancy

```
F_UBii,nfwrot = F_rel × (v(r)² = 4pG?_s·r_s²·[ln(1+x) - x/(1+x)]/r / E_LEP)
× Q_wave × x=r/r_s
Flat for r >> r_s
Source: BB_C_Equations item 1535
```

6.3 SIDM Core Formation Buoyancy

```
F_UBii,sidm = -F_rel × (G = ?·s·v/m / E_LEP) × Q_wave × ln(0.02N)
t_core ˜ (?·s/m)?¹ ˜ 10¹°·(?/108 M_?/kpc³)?¹·(s/m/1 cm²/g)?¹ yr
Core forms when G·t ˜ 1
Source: BB_C_Equations item 1249, 1264
```

---

7. Void and Peculiar Velocity Variants

7.1 Void Density Evolution Buoyancy

```
F_UBii,voidden = -F_rel × (d_v(a) = -3/5·(O_m·a + O_?)^{-3/2}·d_v0 / E_LEP) × Q_wave
a = scale factor, d_v0 = initial underdensity
Integration: d ? a^{-1} in matter domination inside void
Source: BB_C_Equations item 1940, 1338
```

7.2 Peculiar Velocity Buoyancy

```
F_UBii,pec = F_rel × (v_pec = -(fH/3)·?d(r)·r·dr/r² / E_LEP) × Q_wave
f ˜ O_m^{0.55} (growth rate)
Spherical void: integrate Poisson equation
Source: BB_C_Equations item 1341
```

---

8. Convection and Dynamo Variants

8.1 Convective Turnover Time Buoyancy

```
F_UBii,conv = F_rel × (t_conv = H_p/v_conv ; v_conv ˜ (a²gdT·H_p/(4T))^{1/3} / E_LEP) ×
Q_wave
H_p = pressure scale height
a ˜ 2 (mixing length parameter)
Source: BB_C_Equations item 1533
```

8.2 Magnetic Field Reversal Buoyancy (Dynamo Parity)

```
F_UBii,rev = F_rel × (l_rev = (a_dynamo·?)^{1/2}·l_force / E_LEP) × Q_wave
a_dynamo = dynamo a-coefficient (~v/3)
? = magnetic resistivity
Growth vs diffusion sets reversal scale
Source: BB_C_Equations item 1863, 1238
```

8.3 Kazantsev Dynamo Buoyancy

```
F_UBii,dyn = F_rel × (dE_M/dt = (3/2)·E_M/t_eddy·(Re_m^{1/2} - 1) / E_LEP) × Q_wave
Re_m ˜ 10¹° (magnetic Reynolds number in galaxy clusters)
t_eddy = l/v ˜ Myr (eddy turnover for l ~ kpc)
Source: BB_C_Equations item 1234, 1411
```

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9. Star Formation and Feedback Variants

9.1 Metal Enrichment Buoyancy

```
F_UBii,metal = F_rel × (Z = Y·SFR/?_gas - Z·?_out/M_gas / E_LEP) × Q_wave × (Z ˜ 0.1)
Y ˜ 0.02 (stellar yield)
Steady state: Z = Y·SFR/?_out
```

```
Source: BB_C_Equations item 1395, 1571
```

9.2 Photoevaporation (Exoplanet) Buoyancy

```
F_UBii,photo = F_rel × (?_evap = e·L_X·R_p³/(GM_p²·K(?)) / E_LEP) × Q_wave
```

```
L_X ˜ 10²7?²? erg/s (host star X-ray luminosity)
```

```
K(?) = penetration correction factor
```

```
Source: BB_C_Equations item 1496, 1490
```

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10. References

`grok_share_7514fe.txt` lines 2766–2900, 6000–6380 (BB*C*Equations_04Sept2025.pdf Part 2 catalogue)

PAPER*198: F*UBii Taxonomy Part 1 (Compact Objects)

PAPER_196: Triadic Master Equation System
